10-27 第二篇

The Science Behind Weather Forecasts

Will it rain tomorrow? Should we bring an umbrella? We check the weather forecast every day, but few of us stop to think about how it is made. Weather prediction has come a long way from the days when people relied on folk sayings and simple observations. Today, it depends on huge amounts of data, powerful computers, and the work of scientists around the world. [1] Modern weather forecasting is a complex science that combines observation, mathematics, and technology to predict what the atmosphere will do hours or days ahead.

The first step is collecting data. Satellites in space send back images of clouds, storms, and weather systems. Weather stations on the ground measure temperature, humidity, wind speed, and air pressure. Ships, buoys, and airplanes also report conditions from the oceans and the sky. [2] The more data scientists collect from different places and heights, the more accurate their forecasts can become, though the atmosphere is always changing and full of surprises.

Once data is collected, it goes into computer models. These models use the laws of physics to simulate how the atmosphere will evolve. They divide the world into a grid of points and calculate what will happen at each point over time. Supercomputers run these calculations continuously, updating forecasts as new data arrives. [3] Because the atmosphere is chaotic, even small errors in the data can lead to different outcomes, which is why forecasts are less reliable the further ahead we look.

Meteorologists—scientists who study weather—then interpret the results. They compare different models, look at past patterns, and use their experience to fine-tune the forecasts. They also issue warnings when dangerous weather, such as typhoons or heavy snow, is expected. [4] Human expertise remains vital, as computers alone cannot always capture local conditions or explain their forecasts in ways that help people make decisions.

Weather forecasts help us plan our daily lives, but they also save lives. Farmers use them to decide when to plant and harvest. Airlines and shipping companies rely on them for safety. Emergency services use them to prepare for storms and floods. [5] As technology improves and more data becomes available, weather forecasting will continue to get better, helping us stay safe and make smarter choices in a changing climate.

选项

A. Modern weather forecasting is a complex science that combines observation, mathematics, and technology to predict what the atmosphere will do hours or days

ahead. B. The more data scientists collect from different places and heights, the more accurate their forecasts can become, though the atmosphere is always changing and full of surprises. C. Because the atmosphere is chaotic, even small errors in the data can lead to different outcomes, which is why forecasts are less reliable the further ahead we look. D. Human expertise remains vital, as computers alone cannot always capture local conditions or explain their forecasts in ways that help people make decisions. E. As technology improves and more data becomes available, weather forecasting will continue to get better, helping us stay safe and make smarter choices in a changing climate. F. For example, the European Centre for Medium-Range Weather Forecasts runs one of the world's most powerful supercomputers to produce global weather models.

答案键（Answer Key）

[1] A [2] B [3] C [4] D [5] E

深度解析

- **[1] A**：
  - **线索**：空前说预报从谚语和简单观察到依赖数据、计算机和全球科学家，空后需概括现代预报的本质。A的"complex science that combines observation, mathematics, and technology"与空前三大要素对应，"predict what the atmosphere will do hours or days ahead"承接预报目标。
- **[2] B**：
  - **线索**：空前列举多源数据——卫星、地面站、船舶浮标、飞机，空后需说明数据量与预报精度的关系。B的"The more data scientists collect from different places and heights"直接承接前文多源观测，"more accurate... though the atmosphere is always changing"与"full of surprises"合理衔接。
- **[3] C**：
  - **线索**：空前说计算机模型用物理定律模拟大气演变、网格计算、超级计算机持续运行，空后需解释预报的不确定性。C的"Because the atmosphere is chaotic"与模型模拟呼应，"small errors... different outcomes"与"forecasts less reliable the further ahead"形成因果，逻辑完整。
- **[4] D**：
  - **线索**：空前说气象学家解读结果、比较模型、发布预警，空后需强调人的作用。D的"Human expertise remains vital"承接气象学家的工作，"computers alone cannot... capture local conditions or explain"与"interpret、fine-tune、warnings"呼应，突出人机结合。

- [5] E：
    - **线索**：空前说预报在农业、航空、航运、应急中的应用，空后需展望技术发展。E的"As technology improves and more data becomes available"与全文数据和技术主题呼应，"continue to get better... stay safe and make smarter choices"与农业、航空、应急等应用形成收尾。
- **干扰项分析**：
    - **F. For example, the European Centre for Medium-Range Weather Forecasts runs one of the world's most powerful supercomputers to produce global weather models.**：以For example开头，为具体机构举例。各空均为概括或过渡句；[1]或[3]易被误选，但两空分别需概括预报科学与解释不确定性，不需要单个中心案例。